UNIT-2 (Image Transformation & Filtering)

Intensity Transform Functions

Lecture No.-1

Transformation

Transformation is a function. A function that maps one set to another set after performing some operations.

Digital Image Processing system

We have already seen in the introductory tutorials that in digital image processing, we will develop a system that whose input would be an image and output would be an image too. And the system would perform some processing on the input image and gives its output as an processed image. It is shown below.



Now function applied inside this digital system that process an image and convert it into output can be called as transformation function.

As it shows transformation or relation, that how an image1 is converted to image2.

Image transformation.

Consider this equation

$G(x,y) = T\{ f(x,y) \}$

In this equation,

F(x,y) = input image on which transformation function has to be applied.

G(x,y) = the output image or processed image.

T is the transformation function.

This relation between input image and the processed output image can also be represented as.

s = T (r)

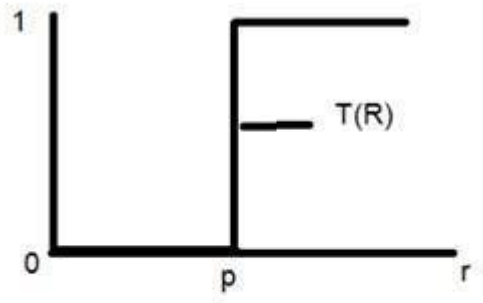
where r is actually the pixel value or gray level intensity of f(x,y) at any point. And s is the pixel value or gray level intensity of g(x,y) at any point.

The basic gray level transformation has been discussed in our tutorial of basic gray level transformations.

Now we are going to discuss some of the very basic transformation functions.

Examples

Consider this transformation function.



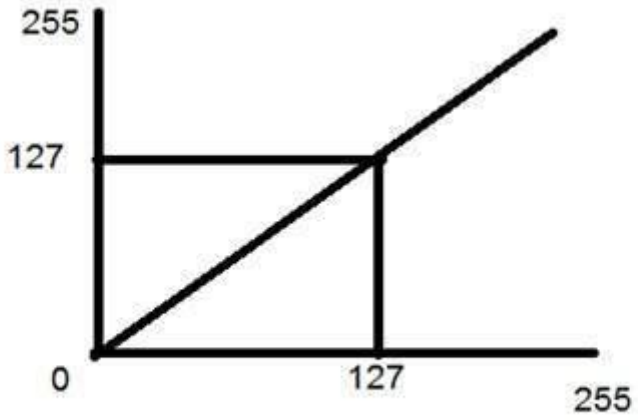
Lets take the point r to be 256, and the point p to be 127. Consider this image to be a one bpp image. That means we have only two levels of intensities that are 0 and 1. So in this case the transformation shown by the graph can be explained as.

All the pixel intensity values that are below 127 (point p) are 0, means black. And all the pixel intensity values that are greater then 127, are 1, that means white. But at the exact point of 127, there is a sudden change in transmission, so we cannot tell that at that exact point, the value would be 0 or 1.

Mathematically this transformation function can be denoted as:

$$g(x,y) = \begin{cases} 0 & f(x,y) < 127 \\ 1 & f(x,y) > 127 \end{cases}$$

Consider another transformation like this



Now if you will look at this particular graph, you will see a straight transition line between input image and output image.

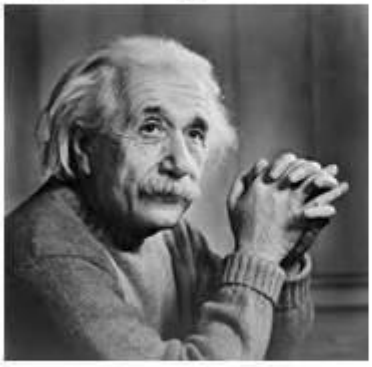
It shows that for each pixel or intensity value of input image, there is a same intensity value of output image. That means the output image is exact replica of the input image.

It can be mathematically represented as:

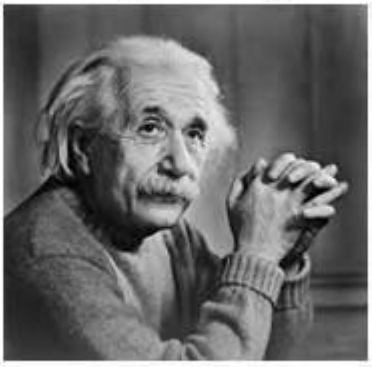
$g(x,y) = f(x,y)$

the input and output image would be in this case are shown below.

Input image



Output image



Intensity transformations are used to increase the contrast between certain intensity values so thatthings can be pick out in an image. In Fig. two images show an image before and

Luminance is a photometric measure of the luminous intensity per unit area of light travelling in a given direction. It describes the amount of light that passes through, is emitted from, or is reflected from a particular area, and falls within a given solid angle.

Gamma Transformations

Gamma Transformations, is used to curve the grayscale components either to brighten the intensity (when gamma is less than one) or darken the intensity (when gamma is greater thanone).

UNIT-2 (Image Transformation & Filtering)

Histogram Processing

Lecture No.-2

Histograms Introduction

In digital image processing, the histogram is used for graphical representation of a digital image. A graph is a plot by the number of pixels for each tonal value. Nowadays, image histogram is present in digital cameras. Photographers use them to see the distribution of tones captured.

In a graph, the horizontal axis of the graph is used to represent tonal variations whereas the vertical axis is used to represent the number of pixels in that particular pixel. Black and dark areas are represented in the left side of the horizontal axis, medium grey color is represented in the middle, and the vertical axis represents the size of the area.





Histogram of the above scenery

Applications of Histograms

1. In digital image processing, histograms are used for simple calculations in software.
2. It is used to analyze an image. Properties of an image can be predicted by the detailed study of the histogram.
3. The brightness of the image can be adjusted by having the details of its histogram.
4. The contrast of the image can be adjusted according to the need by having details of the x-axis of a histogram.
5. It is used for image equalization. Gray level intensities are expanded along the x-axis to produce a high contrast image.
6. Histograms are used in thresholding as it improves the appearance of the image.
7. If we have input and output histogram of an image, we can determine which type of transformation is applied in the algorithm.

Histogram Processing Techniques

Histogram Sliding

In Histogram sliding, the complete histogram is shifted towards rightwards or leftwards. When a histogram is shifted towards the right or left, clear changes are seen in the brightness of the image. The brightness of the image is defined by the intensity of light which is emitted by a particular light source.

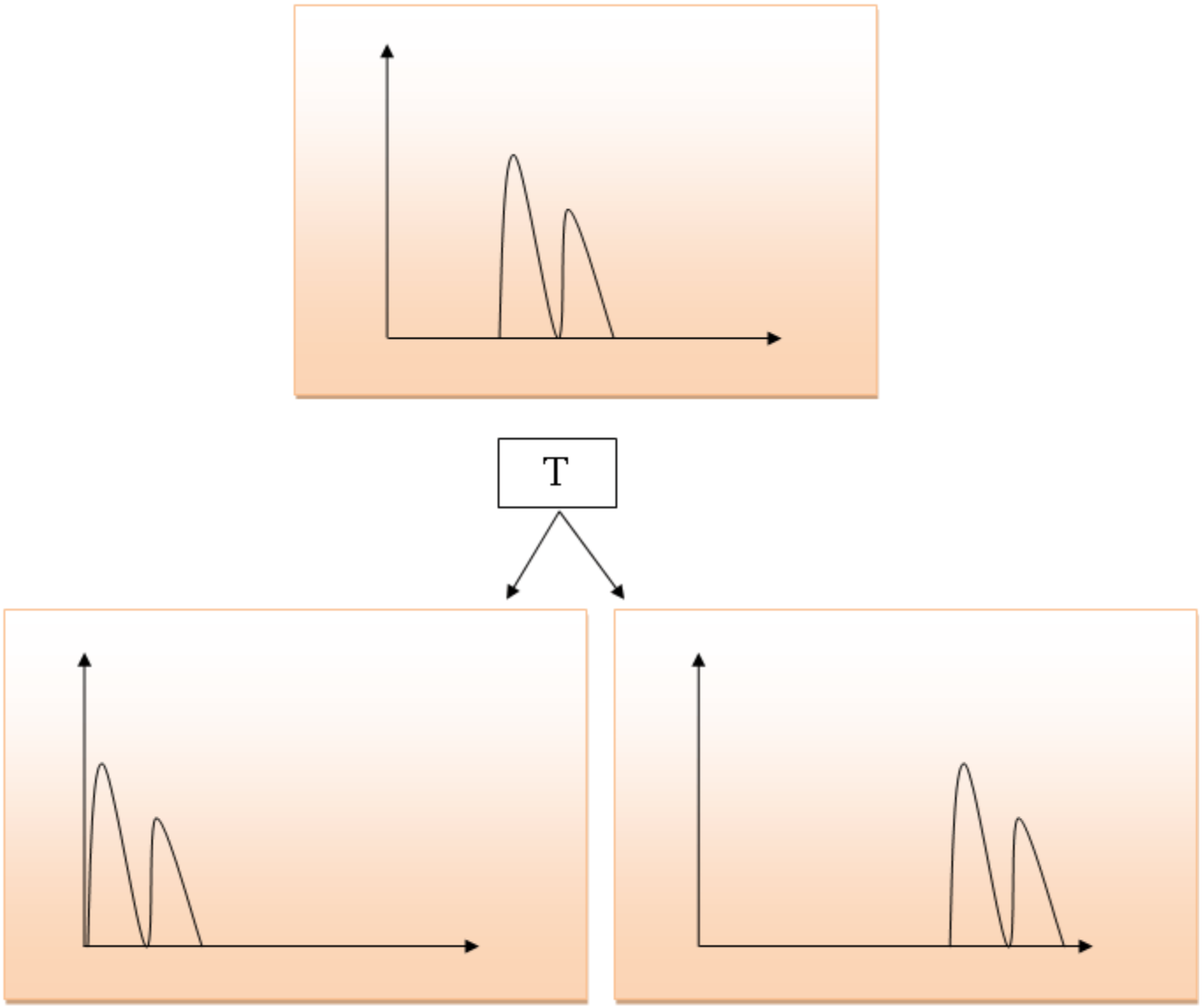


Fig. Histogram Sliding

Histogram Stretching

In histogram stretching, contrast of an image is increased. The contrast of an image is defined between the maximum and minimum value of pixel intensity.

Image processing is the rapidly growing technology presently and it has become an integral part

of the engineering and computer science disciplines.

Histogram of An Image

A histogram of an image represent the graphical interpretation of the image's pixel intensity values.

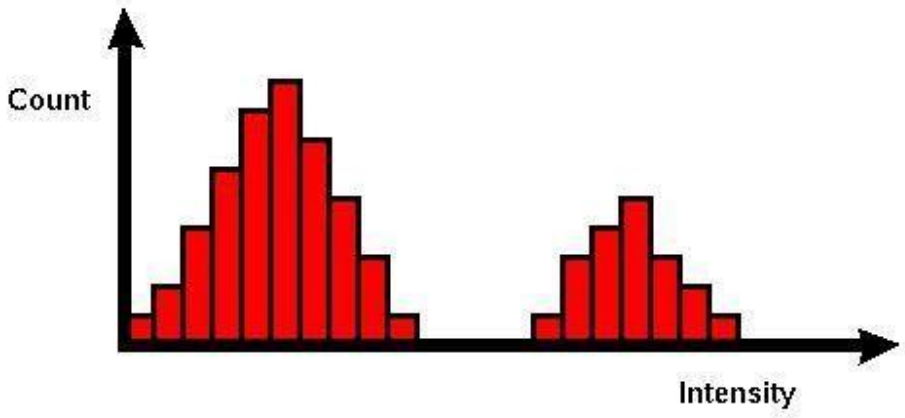


Fig. 2.3 Histogram of an image

As, shown in Fig. 2.3, the X-axis represents the pixel intensity levels of the image and it is usually ranges from 0 to 255. For a gray-scale image, there is only one histogram, while in RGB colored image three 2-D histograms — one for each color are present. The Y-axis of the histogram indicates the frequency or the number of pixels that have specific intensity values.

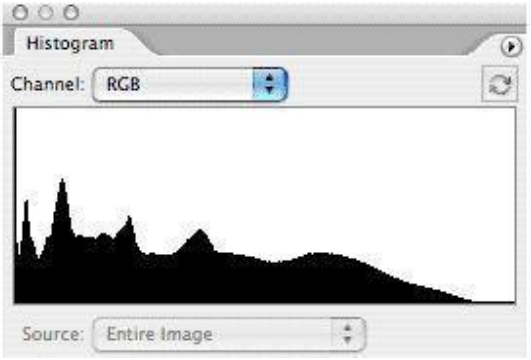


Fig. 2.4 Histogram of gray scale image

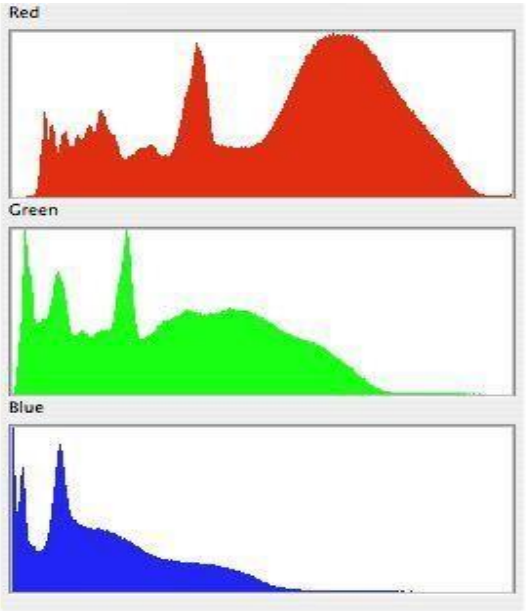


Fig. 2.5 Histogram of an RGB image

Histogram Equalization

It is an image processing technique that adjusts the contrast of an image by using its histogram.

To enhance the image's contrast, the most frequent pixel intensity value are spread out or stretches out the intensity range of the image. This, histogram equalization allows the image's areas with lower contrast to gain a higher contrast.

UNIT-2(Image Transformation & Filtering)

Spatial Filtering

Lecture No.-3

Spatial filtering is a process by which we can alter properties of an optical image by selectively removing certain spatial frequencies that make up an object, for example, filtering video data received from satellite and space probes, or removal of raster from a television picture or scanned image.

Spatial Filtering and its Types

Spatial Filtering technique is used directly on pixels of an image. Mask is usually considered to be added in size so that it has specific center pixel. This mask is moved on the image such that the center of the mask traverses all image pixels.

Classification on the basis of linearity:
There are two types:

1. Linear Spatial Filter
2. Non-linear Spatial Filter

General Classification:

Smoothing Spatial Filter: Smoothing filter is used for blurring and noise reduction in the image. Blurring is pre-processing steps for removal of small details and Noise Reduction is accomplished by blurring.

Types of Smoothing Spatial Filter:

1. Linear Filter (Mean Filter)
2. Order Statistics (Non-linear) filter

These are explained as following below.

1. **Mean Filter:**
   Linear spatial filter is simply the average of the pixels contained in the neighborhood of the filter mask. The idea is replacing the value of every pixel in an image by the average of the grey levels in the neighborhood define by the filter mask.

   Types of Mean filter:

   - **(i) Averaging filter:** It is used in reduction of the detail in image. All coefficients are equal.

- **(ii) Weighted averaging filter:** In this, pixels are multiplied by different coefficients. Center pixel is multiplied by a higher value than average filter.

2. **Order Statistics Filter:**
It is based on the ordering the pixels contained in the image area encompassed by the filter. It replaces the value of the center pixel with the value determined by the ranking result. Edges are better preserved in this filtering.

Types of Order statistics filter:

- **(i) Minimum filter:** 0th percentile filter is the minimum filter. The value of the center is replaced by the smallest value in the window.
- **(ii) Maximum filter:** 100th percentile filter is the maximum filter. The value of the center is replaced by the largest value in the window.
- **(iii) Median filter:** Each pixel in the image is considered. First neighboring pixels are sorted and original values of the pixel is replaced by the median of the list.

Sharpening Spatial Filter: It is also known as derivative filter. The purpose of the sharpening spatial filter is just the opposite of the smoothing spatial filter. Its main focus in on the removal of blurring and highlight the edges. It is based on the first and second order derivative.

First order derivative:

- Must be zero in flat segments.
- Must be non zero at the onset of a grey level step.
- Must be non zero along ramps.

First order derivative in 1-D is given by:

$f' = f(x+1) - f(x)$

Second order derivative:

- Must be zero in flat areas.
- Must be zero at the onset and end of a ramp.
- Must be zero along ramps.

Second order derivative in 1-D is given by:

$f'' = f(x+1) + f(x-1) - 2f(x)$

A General Concept

The **spatial domain enhancement** is based on pixels in a small range (neighbor). This means the transformed intensity is determined by the gray values of those points

within the neighborhood, and thus the spatial domain enhancement is also called **neighborhood operation** or **neighborhood processing**.

A digital image can be viewed as a two-dimensional function f (x, y), and the x-y plane indicates spatial position information, called the **spatial domain**. The filtering operation based on the x-y space neighborhood is called **spatial domain filtering**.

Spatial filtering with a template

The filtering process is to move the filter point-by-point in the image function f (x, y) so that the center of the filter coincides with the point (x, y). At each point (x, y), the filter's response is calculated based on the specific content of the filter and through a predefined relationship called 'template'.

If the pixel in the neighborhood is calculated as a linear operation, it is also called 'linear spatial domain filtering', otherwise, it's called 'nonlinear spatial domain filtering'. Figure 2.3.1 shows the process of spatial filtering with a 3 × 3 template (also known as a filter, kernel, or window).

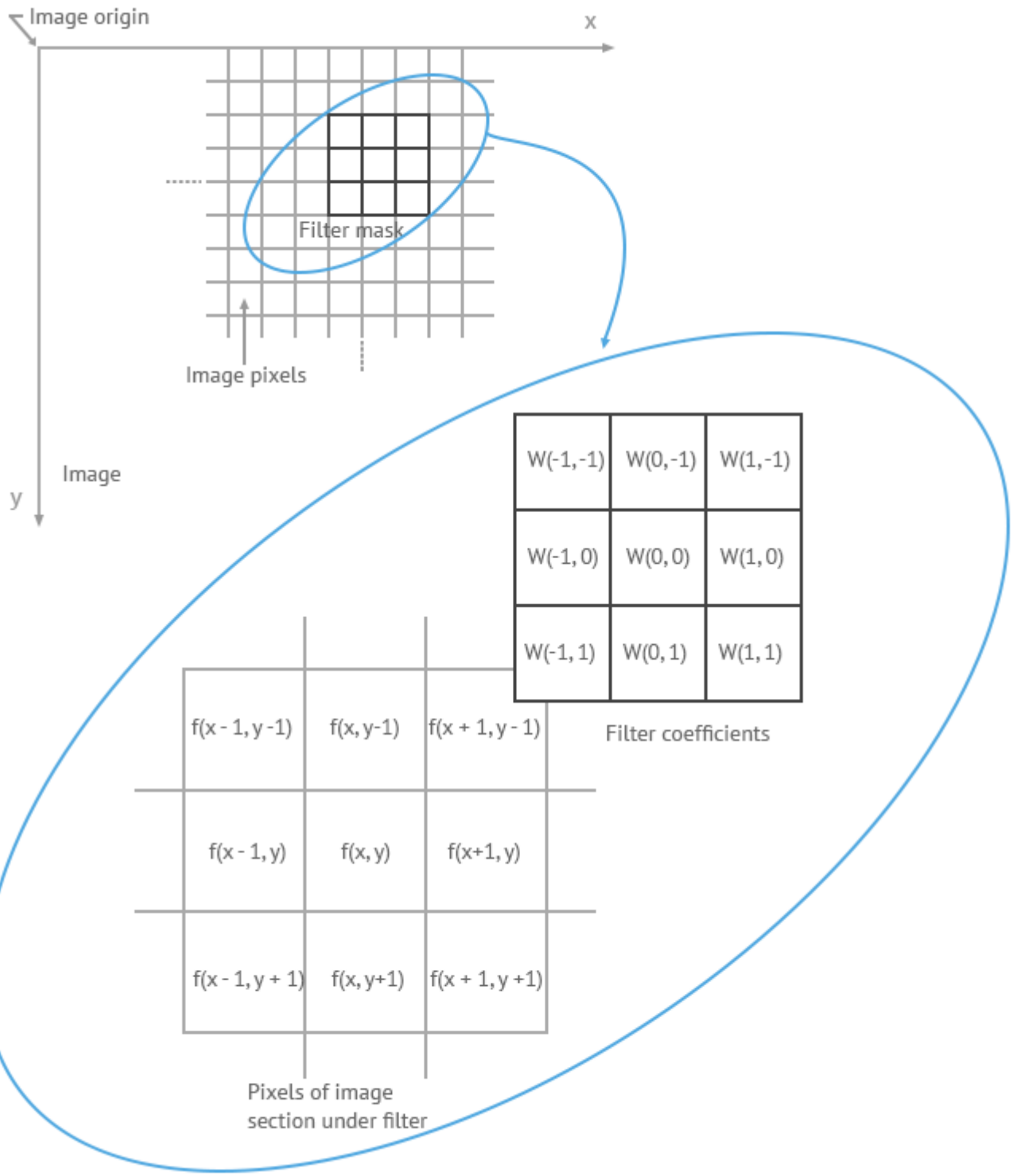


Figure 2.3.1

The coefficients of the filter in linear spatial filtering give a weighting pattern. For

example, for Figure 2.3.1, the response ‘R’ to the template is:

R = w(-1, -1) * f (x-1, y-1) + w(-1, 0) * f (x-1, y) + …+ w(0, 0) * f (x, y) +…+ w(1, 0) * f (x+1, y) + w (1, 1) * f(x+1, y+1)

In mathematics, this is known as element-wise matrix multiplication. For a filter with a size of (2a+1, 2b+1), the output response can be calculated with the following function:

$$g(x,y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t) f(x+s, y+t)$$

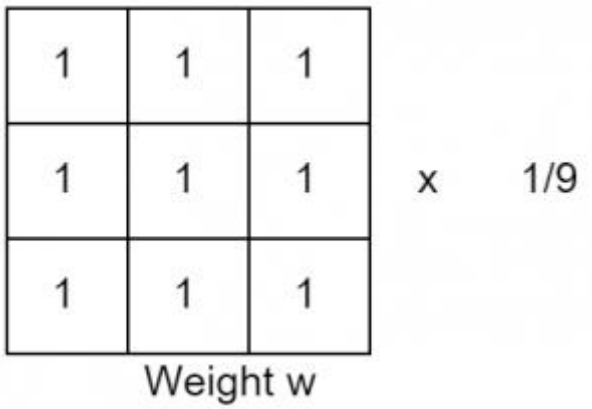
In the following, we will take a look at the filters of image smoothing and sharpening.

Smoothing Filters

Image smoothing is a digital image processing technique that reduces and suppresses image noises. In the spatial domain, neighborhood averaging can generally be used to achieve the purpose of smoothing. Commonly seen smoothing filters include average smoothing, Gaussian smoothing, and adaptive smoothing.

Average Smoothing

First, let’s take a look at the smoothing filter in its simplest form — average template and its implementation.



The points in the 3 × 3 neighborhood centered on the point (x, y) are altogether involved in determining the (x, y) point pixel in the new image ‘g’. All coefficients being 1 means that they contribute the same (weight) in the process of calculating the g(x, y) value. The last coefficient, 1/9, is to ensure that the sum of the entire template elements is 1. This keeps the new image in the same grayscale range as the original image (e.g., [0, 255]). Such a ‘w’ is called an average template.

How it works?

In general, the intensity values of adjacent pixels are similar, and the noise causes grayscale jumps at noise points. However, it is reasonable to assume that occasional noises do not change the local continuity of an image. Take the image below for example, there are two dark points in the bright area.

f_sub=

200	215	212	208	196
198	5	202	199	221
199	207	202	201	211
203	218	210	210	198
203	218	210	0	198
200	215	212	208	205

For the borders, we can add a padding using the “replicate” approach. When smoothing the image with a 3×3 average template, the resulting image is the following.

g_sub=

181	184	186	206	205
180	182	183	206	207
181	183	184	206	208
206	208	186	182	181
207	210	189	183	180
206	209	189	184	181

The two noises are replaced with the average of their surrounding points. The process of reducing the influence of noise is called smoothing or blurring.

Gaussian Smoothing

The average smoothing treats the same to all the pixels in the neighborhood. In order to reduce the blur in the smoothing process and obtain a more natural smoothing effect, it is natural to think to increase the weight of the template center point and reduce the weight of distant points. So that the new center point intensity is closer to its nearest neighbors. The Gaussian template is based on such consideration. The commonly used 3 × 3 Gaussian template is shown below.

w=1/16*

1	2	1
2	4	2
1	2	1

Adaptive Smoothing

The average template blurs the image while eliminating the noise. Gaussian template does a better job, but the blurring is still inevitable as it's rooted in the mechanism. A more desirable way is selective smoothing, that is, smoothing only in the noise area, and not smoothing in the noise-free area. This way potentially minimizes the influence of the blur. It is called **adaptive filtering**.

So how to determine if the local area needs to be smoothed with noise? The answer lies in the nature of the noise, that is, the local continuity. The presence of noise causes a grayscale jump at the noise point, thus making a large grayscale span. Therefore, one of the following two can be used as the criterion:

1. The difference between the maximum intensity and the minimum intensity of a local area is greater than a certain threshold T, ie: max(R) – min(R) > T, where R represents the local area.
2. The variance is greater than a certain threshold T, i.e, D(R) > T, where D(R) represents the variance of the pixels in the area R.

Others

There are some other approaches to tackle the smoothing, such as median filter and adaptive median filter.

Sharpening Filters

Image sharpening filters highlight edges by removing blur. It enhances the grayscale transition of an image, which is the opposite of image smoothing. The arithmetic operators of smoothing and sharpening also testifies the fact. While linear smoothing is based on the weighted summation or integral operation on the neighborhood, the sharpening is based on the derivative (gradient) or finite difference.

How to distinguish noises and edges still matters in sharpening. The difference is that, in smoothing we try to smooth noise and ignore edges and in sharpening we try to enhance edges and ignore noise.

Some applications of where sharpening filters are used include:

- Medical image visualization
- Photo enhancement
- Industrial defect detection
- Autonomous guidance in military systems

There are a couple of filters that can be used for sharpening. One of the most popular filters is Laplace operator. It is based on second order differential.

The corresponding filter template is as follows:

w1=

0	-1	0
-1	4	-1
0	-1	0

With the sharpening enhancement, two numbers with the same absolute value represent the same response, so w1 is equivalent to the following template w2:

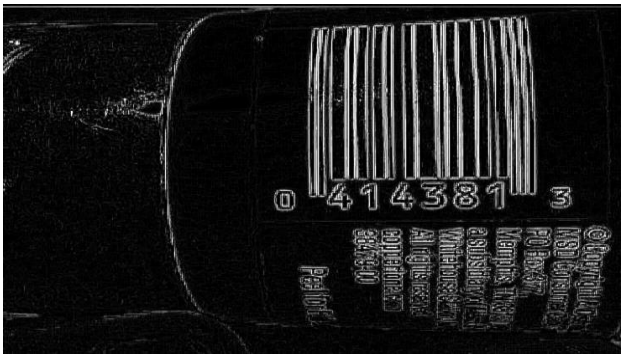
w2=

0	1	0
1	-4	1
0	1	0

Taking a further look at the structure of the Laplacian template, we see that the template is isotropic for a 90-degree rotation. Laplace operator performs well for edges in the horizontal direction and the vertical direction, thus avoiding the hassle of having to filter twice.

Example





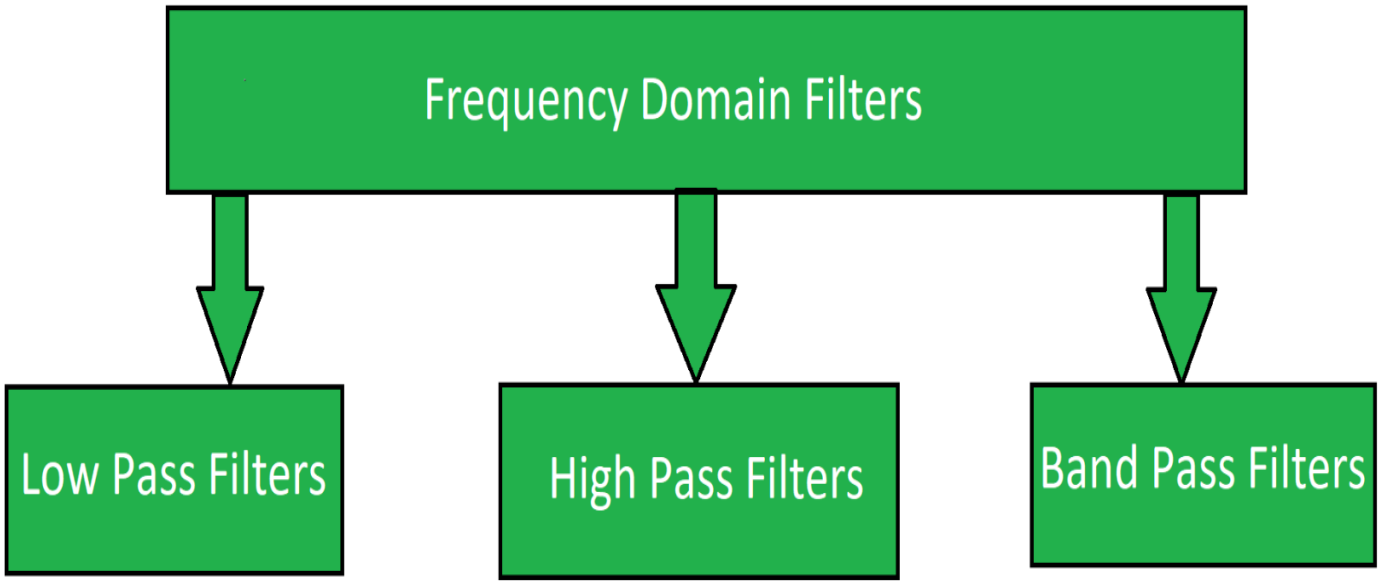
UNIT-2 (Image Transformation & Filtering)

Fourier Transform & Its Properties

Lecture No.-4

Frequency Domain Filters and its Types

Frequency Domain Filters are used for smoothing and sharpening of image by removal of high or low frequency components. Sometimes it is possible of removal of very high and very low frequency. Frequency domain filters are different from spatial domain filters as it basically focuses on the frequency of the images. It is basically done for two basic operation i.e., Smoothing and Sharpening.
These are of 3 types:



Classification of Frequency Domain Filters

1. Low pass filter:
Low pass filter removes the high frequency components that means it keeps low frequency components. It is used for smoothing the image. It is used to smoothen the image by attenuating high frequency components and preserving low frequency components.
Mechanism of low pass filtering in frequency domain is given by:

$$G(u, v) = H(u, v) \cdot F(u, v)$$

where F(u, v) is the Fourier Transform of original image

and H(u, v) is the Fourier Transform of filtering mask

2. High pass filter:
High pass filter removes the low frequency components that means it keeps high frequency components. It is used for sharpening the image. It is used to sharpen the image by attenuating low frequency components and preserving high frequency components.
Mechanism of high pass filtering in frequency domain is given by:
H(u, v) = 1 - H'(u, v)

where H(u, v) is the Fourier Transform of high pass filtering

and H'(u, v) is the Fourier Transform of low pass filtering

3. Band pass filter:
Band pass filter removes the very low frequency and very high frequency components that means it keeps the moderate range band of frequencies. Band pass filtering is used to enhance edges while reducing the noise at the same time.

Fourier Series and Transform

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Next Page

In the last tutorial of Frequency domain analysis, we discussed that Fourier series and Fourier transform are used to convert a signal to frequency domain.

Fourier

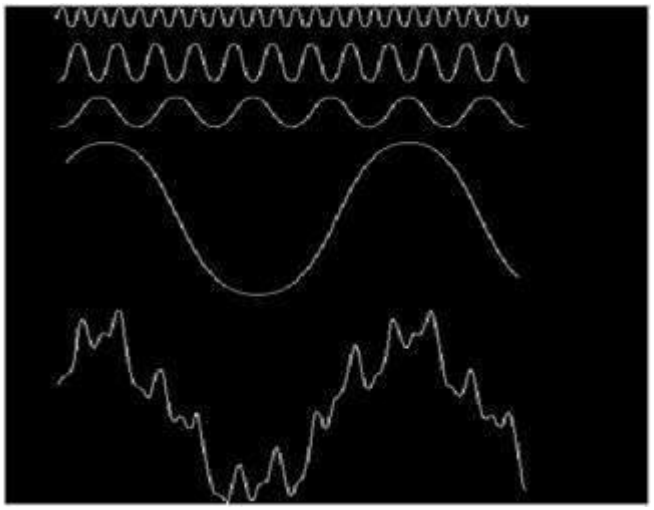
Fourier was a mathematician in 1822. He give Fourier series and Fourier transform to convert a signal into frequency domain.

Fourier Series

Fourier series simply states that , periodic signals can be represented into sum of sines and cosines when multiplied with a certain weight.It further states that periodic signals can be broken down into further signals with the following properties.

- The signals are sines and cosines
- The signals are harmonics of each other

It can be pictorially viewed as



In the above signal , the last signal is actually the sum of all the above signals. This was the idea of the Fourier.

How it is calculated.

Since as we have seen in the frequency domain , that in order to process an image in frequency domain , we need to first convert it using into frequency domain and we have to take inverse of the output to convert it back into spatial domain. That's why both Fourier series and Fourier transform has two formulas. One for conversion and one converting it back to the spatial domain.

Fourier series

The Fourier series can be denoted by this formula.

$$F(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e^{-j2\pi(ux+vy)} \, dx \, d$$

The inverse can be calculated by this formula.

$$f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u, v) e^{j2\pi(ux+vy)} \, du \, dv.$$

Fourier transform

The Fourier transform simply states that that the non periodic signals whose area under the curve is finite can also be represented into integrals of the sines and cosines after being multiplied by a certain weight.

The Fourier transform has many wide applications that include , image compression (e.g JPEG compression) , filtrering and image analysis.

Difference between Fourier series and transform

Although both Fourier series and Fourier transform are given by Fourier , but the difference between them is Fourier series is applied on periodic signals and Fourier transform is applied for non periodic signals

Which one is applied on images.

Now the question is that which one is applied on the images , the Fourier series or the Fourier transform. Well , the answer to this question lies in the fact that what images are. Images are non – periodic. And since the images are non periodic , so Fourier transform is used to convert them into frequency domain.

Discrete fourier transform.

Since we are dealing with images, and infact digital images , so for digital images we will be working on discrete fourier transform



Consider the above Fourier term of a sinusoid. It include three things.

- Spatial Frequency
- Magnitude
- Phase

The spatial frequency directly relates with the brightness of the image. The magnitude of the sinusoid directly relates with the contrast. Contrast is the difference between maximum and minimum pixel intensity. Phase contains the color information.

The formula for 2 dimensional discrete Fourier transform is given below.

$$F(u,v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi(ux/M + vy/N)}$$

The discrete Fourier transform is actually the sampled Fourier transform, so it contains some samples that denotes an image. In the above formula f(x,y) denotes the image , and F(u,v) denotes the discrete Fourier transform. The formula for 2 dimensional inverse discrete Fourier transform is given below.

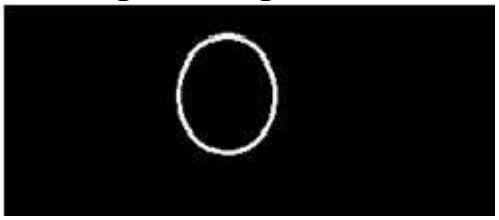
$$f(x,y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u,v) e^{j2\pi(ux/M + vy/N)}$$

The inverse discrete Fourier transform converts the Fourier transform back to the image

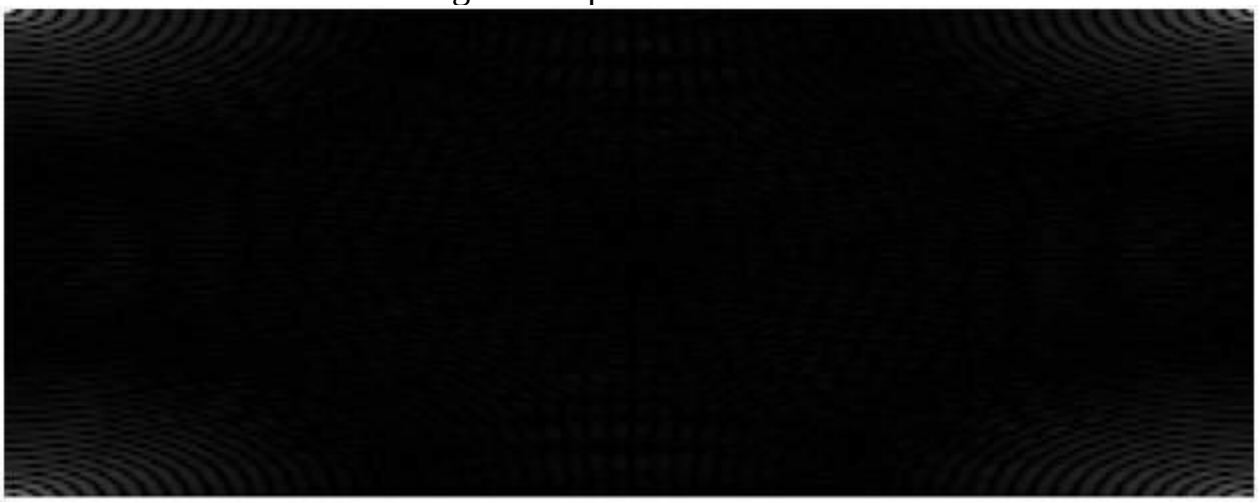
Consider this signal.

Now we will see an image , whose we will calculate FFT magnitude spectrum and then shifted FFT magnitude spectrum and then we will take Log of that shifted spectrum.

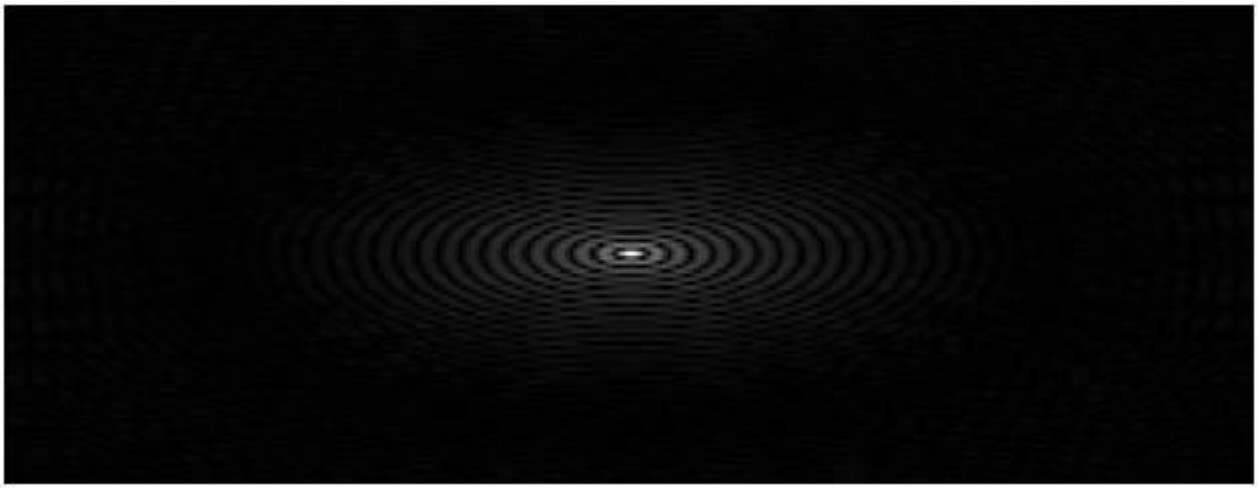
Original Image



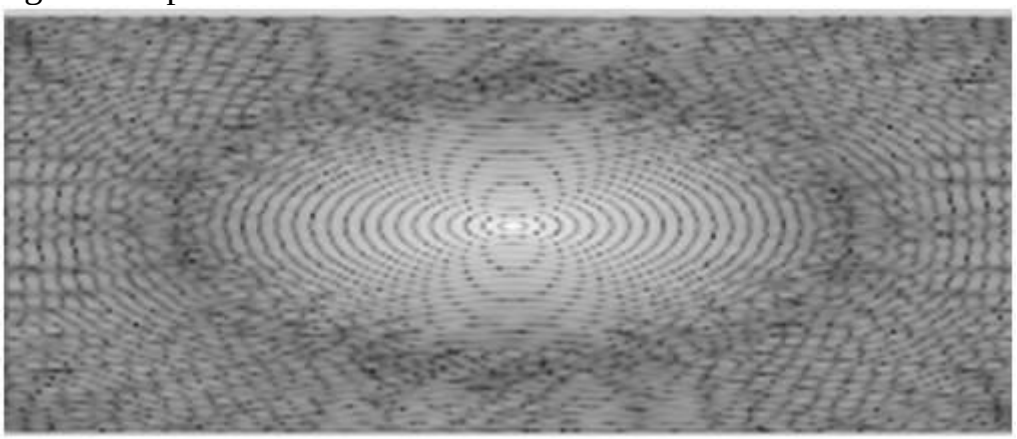
The Fourier transform magnitude spectrum



The Shifted Fourier transform



The Shifted Magnitude Spectrum



Here are the properties of Fourier Transform:

Linearity Property

$$\text{If } x(t) \overset{\text{F.T}}{\longleftrightarrow} X(\omega)$$

$$\& \; y(t) \overset{\text{F.T}}{\longleftrightarrow} Y(\omega)$$

Then linearity property states that

$$ax(t) + by(t) \overset{\text{F.T}}{\longleftrightarrow} aX(\omega) + bY(\omega)$$

Time Shifting Property

$$\text{If} x(t) \overset{\text{F.T}}{\longleftrightarrow} X(\omega)$$

Then Time shifting property states that

$$x(t - t_0) \overset{\text{F.T}}{\longleftrightarrow} e^{-j\omega t_0} X(\omega)$$

Frequency Shifting Property

$$\text{If } x(t) \overset{\text{F.T}}{\longleftrightarrow} X(\omega)$$

Then frequency shifting property states that

$$e^{j\omega_0 t}.\, x(t) \overset{\text{F.T}}{\longleftrightarrow} X(\omega - \omega_0)$$

Time Reversal Property

$$\text{If } x(t) \overset{\text{F.T}}{\longleftrightarrow} X(\omega)$$

Then Time reversal property states that

$$x(-t) \overset{\text{F.T}}{\longleftrightarrow} X(-\omega)$$

Time Scaling Property

$$\text{If } x(t) \overset{\text{F.T}}{\longleftrightarrow} X(\omega)$$

$$\text{If } x(t) \xleftrightarrow{\text{F.T}} X(\omega)$$

Then Time scaling property states that

$$x(at) \frac{1}{|a|} X\frac{\omega}{a}$$

Differentiation and Integration Properties

$$If\ x(t) \xleftrightarrow{\text{F.T}} X(\omega)$$

Then Differentiation property states that

$$\frac{dx(t)}{dt} \xleftrightarrow{\text{F.T}} j\omega . X(\omega)$$

$$\frac{d^n x(t)}{dt^n} \xleftrightarrow{\text{F.T}} (j\omega)^n . X(\omega)$$

and integration property states that

$$\int x(t)\, dt \xleftrightarrow{\text{F.T}} \frac{1}{j\omega} X(\omega)$$

$$\iiint \dots \int x(t)\,dt \overset{\text{F.T}}{\longleftrightarrow} \frac{1}{(j\omega)^n} X(\omega)$$

Multiplication and Convolution Properties

$$\text{If } x(t) \overset{\text{F.T}}{\longleftrightarrow} X(\omega)$$

$$\& \; y(t) \overset{\text{F.T}}{\longleftrightarrow} Y(\omega)$$

Then multiplication property states that

$$x(t).y(t) \overset{\text{F.T}}{\longleftrightarrow} X(\omega) * Y(\omega)$$

and convolution property states that

$$x(t) * y(t) \overset{\text{F.T}}{\longleftrightarrow} \frac{1}{2\pi} X(\omega).Y(\omega)$$

Transformation functions



This method is known as transformations, in which we discussed different type of transformations and some gray level transformations

Another way of dealing images

Here we are going to discuss another method of dealing with images. This other method is known as convolution. Usually the black box(system) used for image processing is an LTI system or linear time invariant system. By linear we mean that such a system where output is always linear , neither log nor exponent or any other. And by time invariant we means that a system which remains same during time.

So now we are going to use this third method. It can be represented as.



Convolution is a simple mathematical operation which is fundamental to many

common image processing operators. Convolution provides a way of `multiplying together' two arrays of numbers, generally of different sizes, but of the same dimensionality, to produce a third array of numbers of the same dimensionality.

It can be mathematically represented as two ways

g(x,y) = h(x,y) * f(x,y)

It can be explained as the "mask convolved with an image".

Or

g(x,y) = f(x,y) * h(x,y)

It can be explained as "image convolved with mask".

There are two ways to represent this because the convolution operator(*) is commutative. The h(x,y) is the mask or filter.

What is mask?

Mask is also a signal. It can be represented by a two dimensional matrix. The mask is usually of the order of 1x1, 3x3, 5x5, 7x7 . A mask should always be in odd number, because other wise you cannot find the mid of the mask. Why do we need to find the mid of the mask. The answer lies below, in topic of, how to perform convolution?

How to perform convolution?

In order to perform convolution on an image, following steps should be taken.

- **Flip the mask (horizontally and vertically) only once**
- **Slide the mask onto the image.**
- **Multiply the corresponding elements and then add them**
- **Repeat this procedure until all values of the image has been calculated.**

Example of convolution

Let's perform some convolution. Step 1 is to flip the mask.

Mask

Let's take our mask to be this.

1	**2**	**3**
4	**5**	**6**
7	**8**	**9**

Flipping the mask horizontally

3	**2**	**1**
6	**5**	**4**
9	**8**	**7**

Flipping the mask vertically

9	**8**	**7**
6	**5**	**4**
3	**2**	**1**

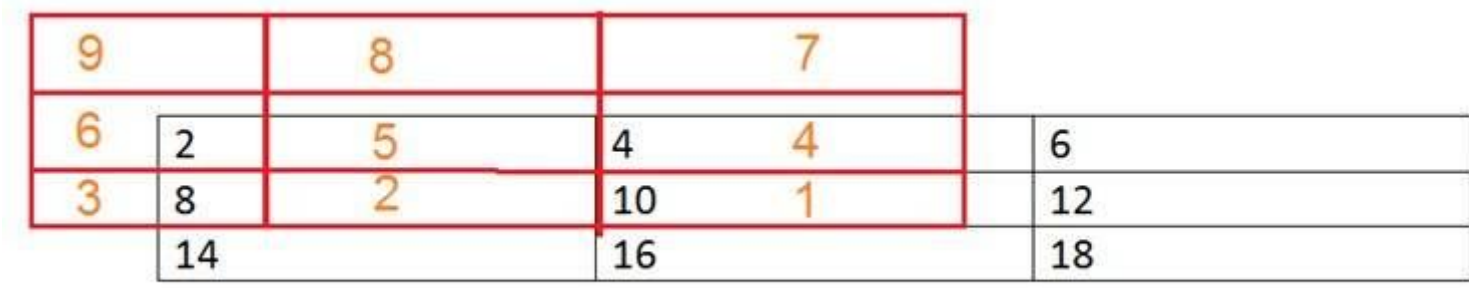
Image

Let's consider an image to be like this

2	**4**	**6**
8	**10**	**12**
14	**16**	**18**

Convolution

Convolving mask over image. It is done in this way. Place the center of the mask at each element of an image. Multiply the corresponding elements and then add them , and paste the result onto the element of the image on which you place the center of mask.



The box in red color is the mask, and the values in the orange are the values of the mask. The black color box and values belong to the image. Now for the first pixel of the image, the value will be calculated as

First pixel = (5*2) + (4*4) + (2*8) + (1*10)

= 10 + 16 + 16 + 10

= 52

Place 52 in the original image at the first index and repeat this procedure for each pixel of the image.

Why Convolution

Convolution can achieve something, that the previous two methods of manipulating images can't achieve. Those include the blurring, sharpening, edge detection, noise reduction e.t.c.

UNIT-2 (Image Transformation & Filtering)
Frequency Domain Filters
Lecture No.-5

Frequency Domain Filters are used for smoothing and sharpening of image by removal of high or low frequency components. Frequency domain filters are different from spatial domain filters as it basically focuses on the frequency of the images. It is basically done for two basic operation i.e., Smoothing and Sharpening.

Whereas convolution is used to evaluate filters in the spatial domain, filtering is implemented in the frequency domain by multiplication. The image in the frequency domain is multiplied by the filter's frequency response in the frequency domain. Thus, the filter transfer function is simply a matrix of the same size as the image. The complex values of the image in the frequency domain are simply multiplied element by element with the filter transfer function to amplify or attenuate specific frequencies of the image. The inverse Fourier transform is then used to generate the output image.

Gaussian smoothing

The Gaussian smoothing operator is a 2-D convolution operator that is used to `blur' images and remove detail and noise. In this sense it is similar to the mean filter, but it uses a different kernel that represents the shape of a Gaussian (`bell-shaped') hump. This kernel has some special properties.

UNIT-2 (Image Transformation & Filtering)Colour Models
Lecture No.-6

A color model is a system used to describe a color. For example with Red, Green and Blue (RGB) elements or Cyan, Magenta, Yellow and Black (CMYK). A color space is a way of mapping real colors to the color model's particular values.

Different color models

Additive and subtractive color models

- RYB color model.
- RGB color model.
- CMY and CMYK color models.
- Background.
- HSL and HSV.
- Munsell color system.
- Natural Color System.
- Preucil hue circle.

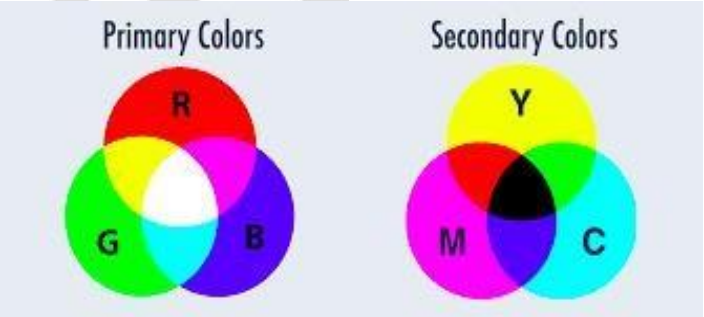


Fig. 2.10Primary & secondary colours

Three popular color models are:

[1] CMYK (Cyan, Magenta, Yellow, Black)

[2] RGB (Red, Green, Blue)

[3] Lab Color

[1] CMYK (Cyan, Magenta, Yellow, Black)

The CMYK model is used for print work and it describes colors based on their percentage of Cyan, Magenta, Yellow and Black. These four colors are used by commercial printers and bureaus and you may also find that your home printer uses these colors too. These four colors are needed to reproduce full color artwork in magazines, books and brochures. By combining Cyan, Magenta, Yellow and Black on paper in varying percentages, the illusion of lots of colors is created.

[2] RGB (Red, Green, Blue)

The RGB model is used when working with screen based designs. A value between 0 and 255 is assigned to each of the light colors, Red, Green and Blue. So for example, if you wanted to create a purely blue color, Red would have a value of 0, Green would have a value of 0 and Blue would have a value of 255 (pure blue). To create black, Red, Green and Blue would each have a value of 0 and to create white, each would have a value of 255. RGB is known as an "additive" model and is the opposite of the subtractive color model.

[3] Lab Color

The Lab color model is a slightly more complex beast. It is made up of three components – the lightness component (L) ranging from 0 to 100, the "a" component comes from the green-red axis in the Adobe Color Picker, and the "b" component which comes from the blue-yellow axis in the Adobe Color Picker. Both "a" and "b" can range from +127 to –128. When Photoshop is converting from one model to another, it uses Lab as the intermediate color

model.

UNIT-2 (Image Transformation & Filtering)

Pseudo Colouring

Lecture No.-7

True color VS false color

True-color images are images with natural color rendition: a red apple appears red, a blue ocean appears blue, etc. It's difficult to achieve absolute true color in images, but being approximately close to human perception is acceptable.

False color images, on the other hand, sacrifices natural color rendition in order to facilitate the detection of some objects. They are mainly used for satellite and space images.

Pseudocolor images are originally grayscale which are assigned colors based on the intensity values. Typical usage of these images is for thermography where the only available is infrared radiation instead of lights. Another example is elevation map.

Intensity Slicing

This is a simple case of pseudocolor image processing. It's also called density slicing or **color**
coding.
Imagine a grayscale image as a 3D function with the intensity being the third dimension. Placing a plane parallel to the horizontal plane for the pixel position coordinates would *"slice"* the image into two parts. After that we can assign different colors to different levels.

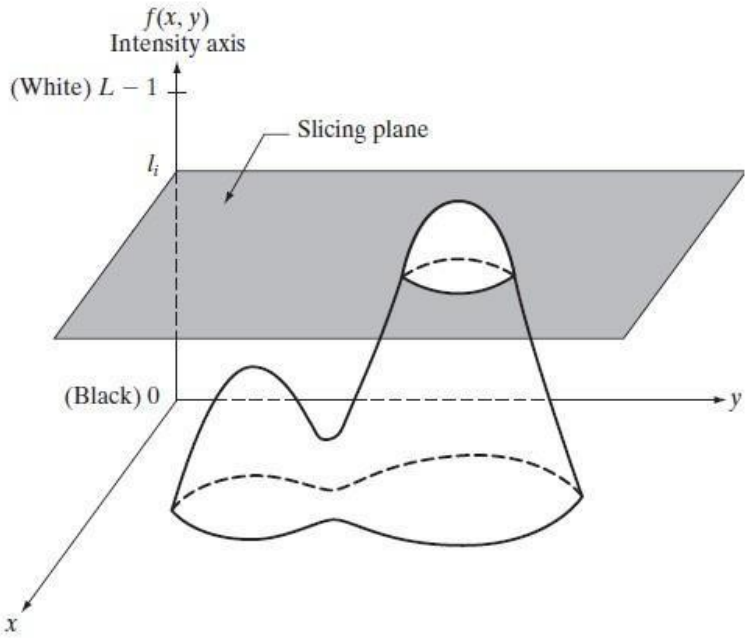


Fig. 2.10 Geometric interpretation of the intensity slicing technique.

We are not constrained to use only one plane. Multiple slices result in more flexible representations of the grayscale images. In this case, the grayscale is parted into intervals andeach one is assigned a different color.

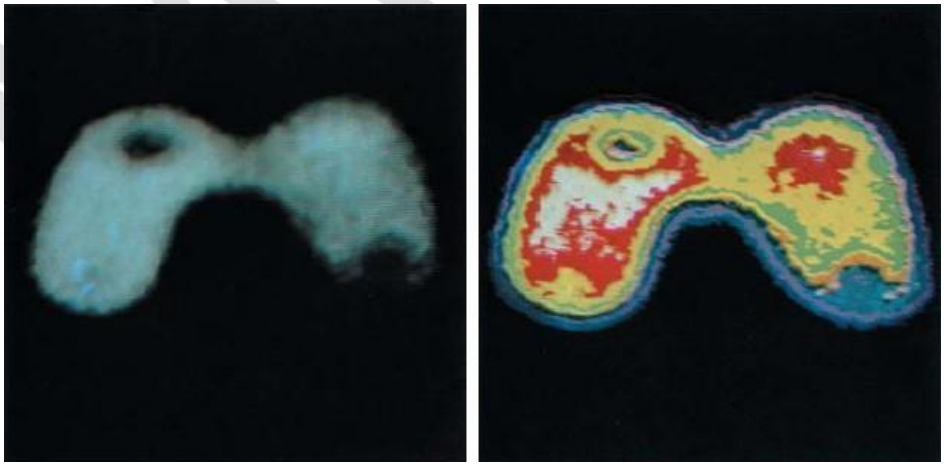


Fig. 2.11Result of density slicing into eight colors

UNIT-2 (Image Transformation & Filtering)

Colour Transform

Lecture No.-8

Color transformation, Color transformation, or color space conversion, is the transformation of the representation of a color from one color space to another. This calculation is required whenever data is exchanged inside a color-managed chain and carried out by a Color Matching Module.

Use Color Transforms to convert three-band red, green, blue (RGB) images to one of several specific color spaces and from the selected color space back to RGB. Adjusting the contrast stretch between the two transforms, you can produce a color-enhanced color composite image.

RGB to CMY conversion

The conversion from RGB to CMY is done using this method.

Consider you have an color image , means you have three different arrays of RED, GREEN and BLUE. Now if you want to convert it into CMY, here's what you have to do. You have to subtract it by the maximum number of levels – 1.

Each matrix is subtracted and its respective CMY matrix is filled with result.

RGB Color Space

$$I = \frac{1}{3}[R + G + B]$$

To modify intens ltiply each color component by k

= [+ +] $I' = [KR + KG + KB] = \frac{K}{3}[R + G + B] = KI$
3

UNIT-2 (Image Transformation & Filtering)

Basics of Wavelet Transform

Lecture No.-9

Wavelet Transformation

A **wavelet** is a wave-like oscillation with an amplitude that begins at zero, increases, and then decreases back to zero. It can typically be visualized as a "brief oscillation" like one recorded by a seismograph or heart monitor.

Wavelets are functions that are concentrated in time and frequency around a certain point. This transformation technique is used to overcome the drawbacks of fourier method. Fourier transformation, although it deals with frequencies, does not provide temporal details. According to *Heisenberg's Uncertainty Principle*, we can either have high frequency resolution and poor temporal resolution or vice versa.

This wavelet transform finds its most appropriate use in non-stationary signals. This transformation achieves good frequency resolution for low-frequency components and high temporal resolution for high-frequency components.

This method starts with a mother wavelet such as Haar, Morlet, Daubechies, etc. The signal is then essentially translated into scaled and shifted versions of mother wavelet.

Wavelet Analysis in Image Processing

Wavelet analysis is used to divide information present on an image (signals) into two discrete components — approximations and details (sub-signals).

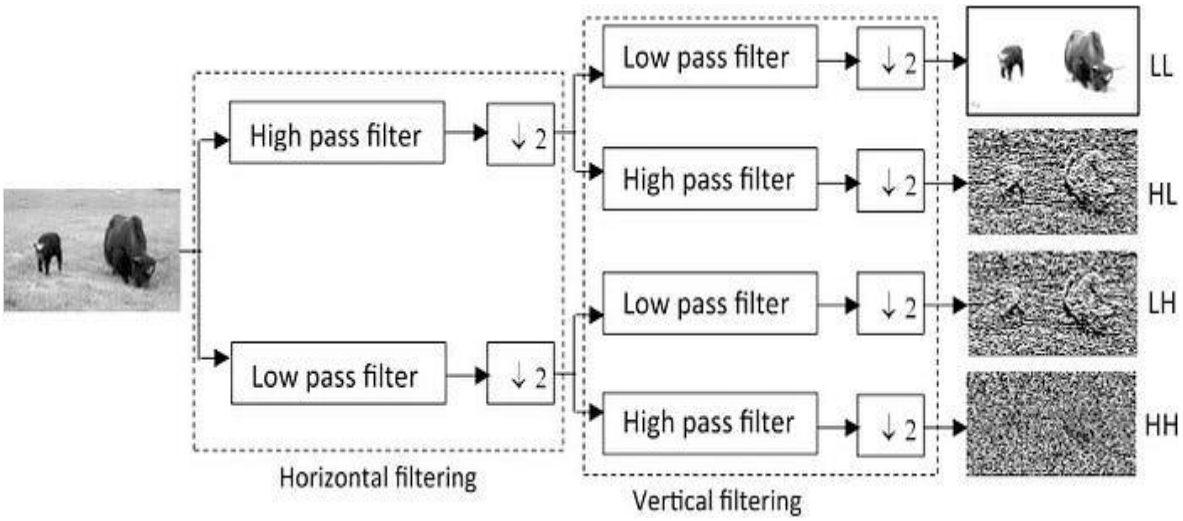


Fig - wavelet based filtering

A wavelet is a mathematical function used to divide a given function or continuous-time signalinto different scale components.

In principle the continuous wavelet transform works by using directly the definition of the wavelet transform, i.e. we are computing a convolution of the signal with the scaled wavelet. For each scale we obtain by this way an array of the same length N as the signal has.

A major disadvantage of the Fourier Transform is it captures *global* frequency information, meaning frequencies that persist over an entire signal. This kind of signal decomposition may not serve all applications well, for example Electrocardiography (ECG) where signals have short intervals of characteristic oscillation. An alternative approach is the **Wavelet Transform**, which decomposes a function into a set of wavelets**.**

A **Wavelet** is a **wave-like oscillation that is localized in time**, an example is given below. Wavelets have two basic properties: scale and location. **Scale** (or dilation) defines how "stretched" or "squished" a wavelet is. This property is related to frequency as defined for waves. **Location** defines where the wavelet is positioned in time (or space).

Continuous Wavelet Transform

- Define the continuous wavelet transform of *f(x)*:

$$W_{\varphi}(s,\tau) = \int_{-\infty}^{\infty} f(x)\psi_{s,\tau}(x)\,dx$$

- This transforms a continuous function of one variable into a continuous function of two variables: translation and scale
- The wavelet coefficients measure how closely correlated the wavelet is with each section of the signal
- For compact representation, choose a wavelet that matches the shape of the image components
  - Example: Haar wavelet for black and white drawings

Example

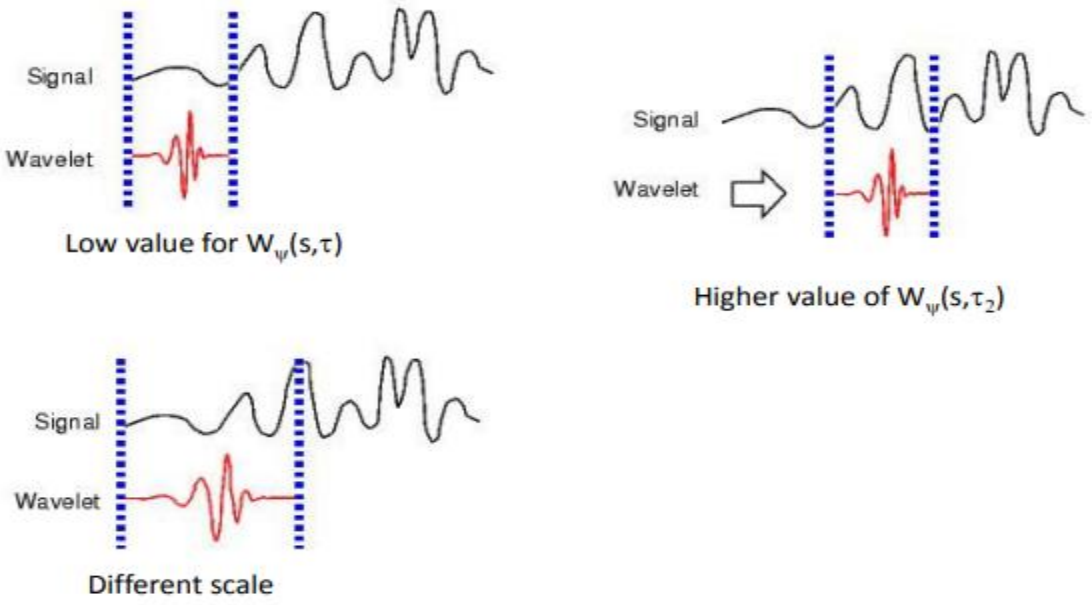


Fig: wavelet and scaling function

References:

1. R. Gonjalez, Digital Image Processing, Third Edition, Pearson
2. S. Shridhar , Digital Image Processing,